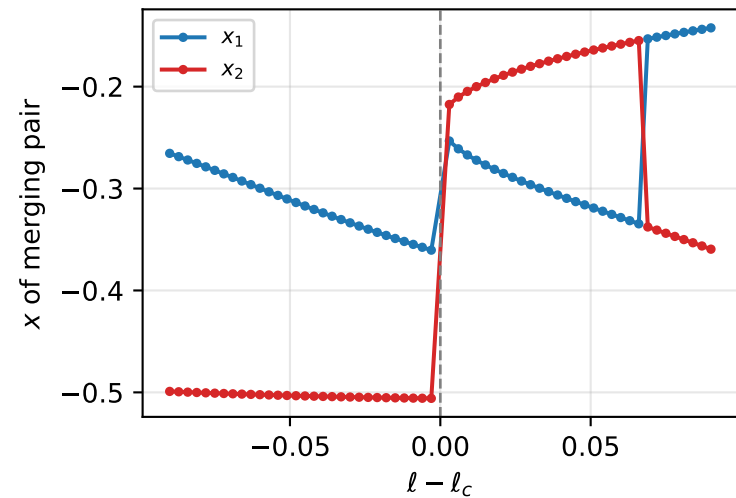
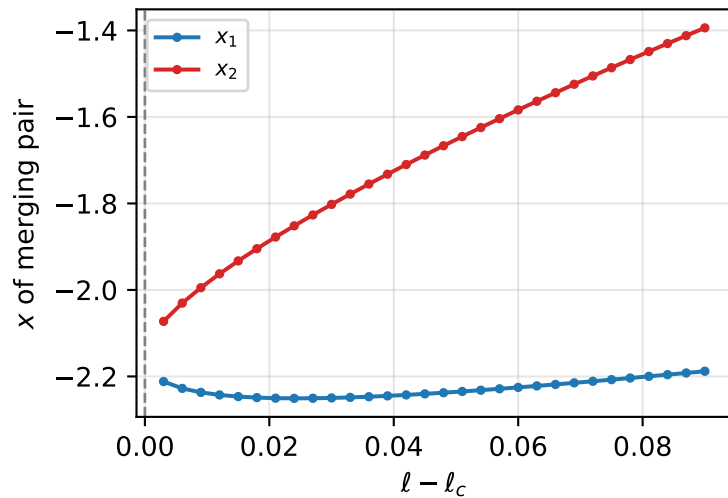


Representative saddle-node folds: the merging pair meets at $\ell = \ell_c$

Case-1 fold ($5 \rightarrow 7$)



Case-2 fold ($7 \rightarrow 9$)



$N = 11$ -bounding fold ($9 \rightarrow 11$)

